

Track 2 Best Model Collage Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report compares representative Track 2 TE-curve predictions for the current best reference, Track 1, Wave 1 directional, and Wave 1 global models. Each model is shown as one four-image collage so local oscillation tracking can be inspected directly.

Scope

- each collage contains four deterministic held-out test curves;
- forward models are shown on forward curves only;
- backward models are shown on backward curves only;
- global Wave 1 models are shown on two forward and two backward curves;
- Measured TE uses the same line width as predictions and a dark-gray color for balanced visual comparison.

Metrics Summary

Forward Reference Best Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_original_best_Fw	rcim_original	Fw	0.002769	0.002951	6.250
paper_retuned_best_Fw	rcim_retuned	Fw	0.001839	0.002041	4.109
track1_best_Fw	rcim_track1	Fw	0.003014	0.003204	6.819

Forward Wave 1 Family Best Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
feedforward_fw	wave1_current_registry	Fw	0.003404	0.003855	7.551
harmonic_regression_fw	wave1_current_registry	Fw	0.003230	0.003494	7.185
periodic_mlp_fw	wave1_current_registry	Fw	0.003254	0.003553	7.232
residual_harmonic_mlp_fw	wave1_current_registry	Fw	0.003273	0.003563	7.266
tree_fw	wave1_current_registry	Fw	0.003053	0.003395	6.731
periodic_mlp_harmonic_fw	wave1_periodic_mlp_harmonic_campaign	Fw	0.003254	0.003553	7.232

Backward Reference Best Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.003675	0.004284	7.572
track1_best_Bw	rcim_track1	Bw	0.005027	0.005212	11.860

Backward Wave 1 Family Best Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
feedforward_bw	wave1_current_registry	Bw	0.003586	0.004023	7.832
harmonic_regression_bw	wave1_current_registry	Bw	0.003678	0.004012	8.058
periodic_mlp_bw	wave1_current_registry	Bw	0.003574	0.004006	7.807
residual_harmonic_mlp_bw	wave1_current_registry	Bw	0.003536	0.003874	7.728
tree_bw	wave1_current_registry	Bw	0.003258	0.003651	7.051
periodic_mlp_harmonic_bw	wave1_periodic_mlp_harmonic_campaign	Bw	0.003583	0.003925	7.875

Forward Wave 2 Temporal Family Best Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
temporal_convolution_fw	wave2_temporal_entry_registry	Fw	0.003603	0.004031	8.028
gru_sequence_fw	wave2_temporal_entry_registry	Fw	0.003330	0.003762	7.378
lstm_sequence_fw	wave2_temporal_entry_registry	Fw	0.003366	0.003800	7.450
periodic_temporal_convolution_fw	wave2_temporal_entry_registry	Fw	0.003335	0.003708	7.404
periodic_gru_sequence_fw	wave2_temporal_entry_registry	Fw	0.003186	0.003438	7.077
periodic_lstm_sequence_fw	wave2_temporal_entry_registry	Fw	0.003266	0.003550	7.258

Backward Wave 2 Temporal Family Best Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
temporal_convolution_bw	wave2_temporal_entry_registry	Bw	0.003742	0.004166	8.184
gru_sequence_bw	wave2_temporal_entry_registry	Bw	0.003626	0.004082	7.907
lstm_sequence_bw	wave2_temporal_entry_registry	Bw	0.003555	0.003985	7.767
periodic_temporal_convolution_bw	wave2_temporal_entry_registry	Bw	0.003628	0.003987	7.979
periodic_gru_sequence_bw	wave2_temporal_entry_registry	Bw	0.002392	0.002639	5.466
periodic_lstm_sequence_bw	wave2_temporal_entry_registry	Bw	0.002625	0.002877	6.013

Global Wave 1 Family Best Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
feedforward_global	wave1_current_registry	global	0.003465	0.003897	7.636
harmonic_regression_global	wave1_current_registry	global	0.018129	0.018330	41.458
periodic_mlp_global	wave1_current_registry	global	0.003447	0.003872	7.582
residual_harmonic_mlp_global	wave1_current_registry	global	0.003407	0.003822	7.486
tree_global	wave1_current_registry	global	0.003144	0.003533	6.854
periodic_mlp_harmonic_global	wave1_periodic_mlp_harmonic_campaign	global	0.003516	0.003810	7.779

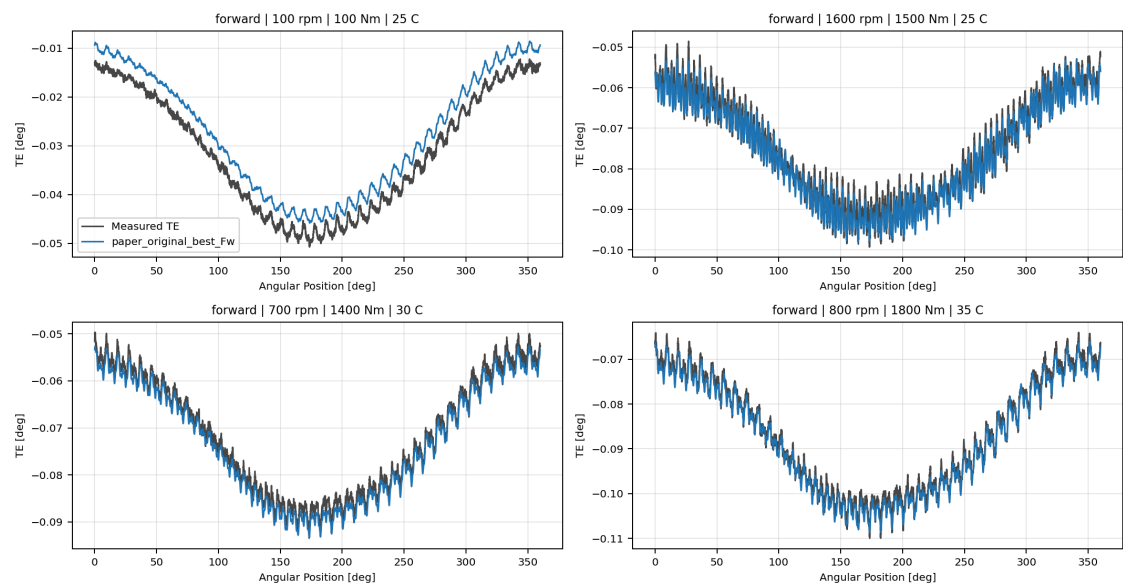
Global Wave 2 Temporal Family Best Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
temporal_convolution_global	wave2_temporal_entry_registry	global	0.003751	0.004183	8.295
gru_sequence_global	wave2_temporal_entry_registry	global	0.003591	0.004028	7.907
lstm_sequence_global	wave2_temporal_entry_registry	global	0.003480	0.003903	7.654
periodic_temporal_convolution_global	wave2_temporal_entry_registry	global	0.003506	0.003836	7.758
periodic_gru_sequence_global	wave2_temporal_entry_registry	global	0.002704	0.002949	6.139
periodic_lstm_sequence_global	wave2_temporal_entry_registry	global	0.002707	0.002958	6.120

Collage Gallery - Forward Reference Best Models

paper_original_best_Fw

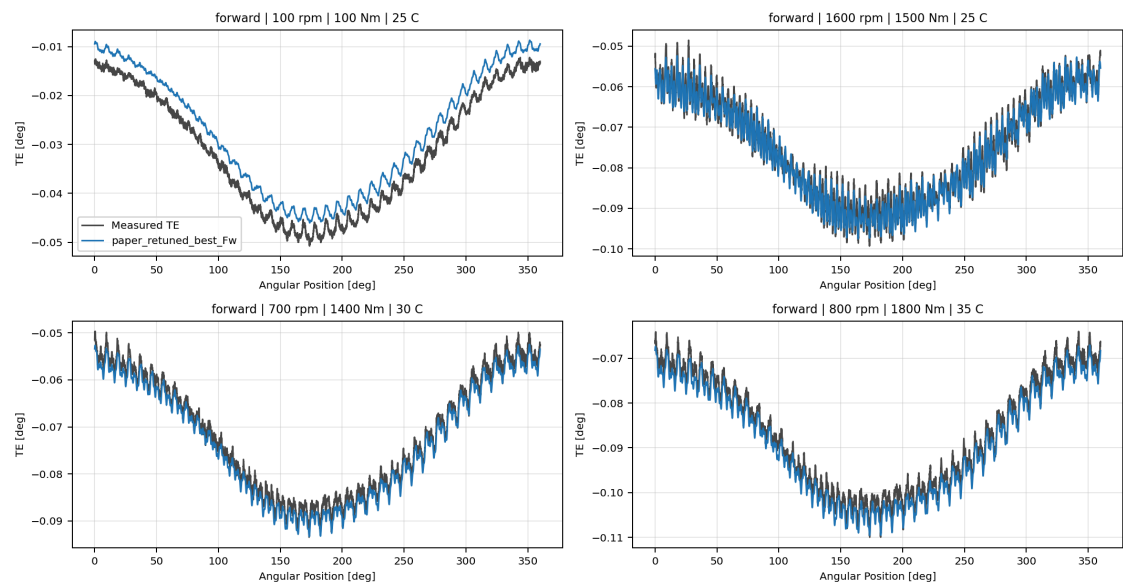
paper_original_best_Fw



paper_original_best_Fw Track 2 collage

paper_retuned_best_Fw

paper_retuned_best_Fw

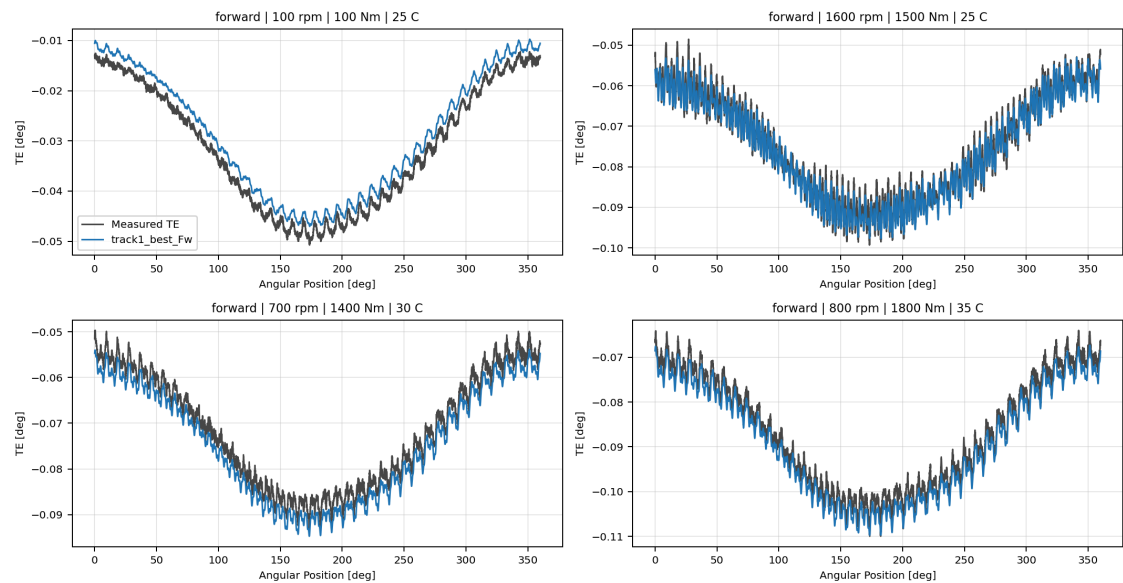


paper_retuned_best_Fw Track 2 collage

Collage Gallery - Forward Reference Best Models Continued

track1_best_Fw

track1_best_Fw

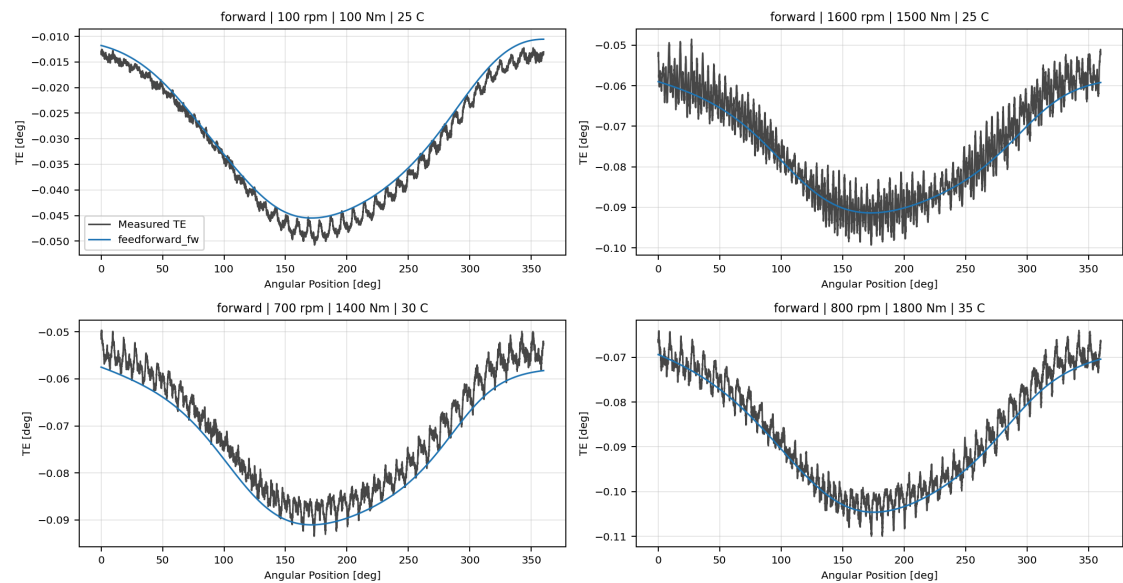


track1_best_Fw Track 2 collage

Collage Gallery - Forward Wave 1 Family Best Models

feedforward_fw

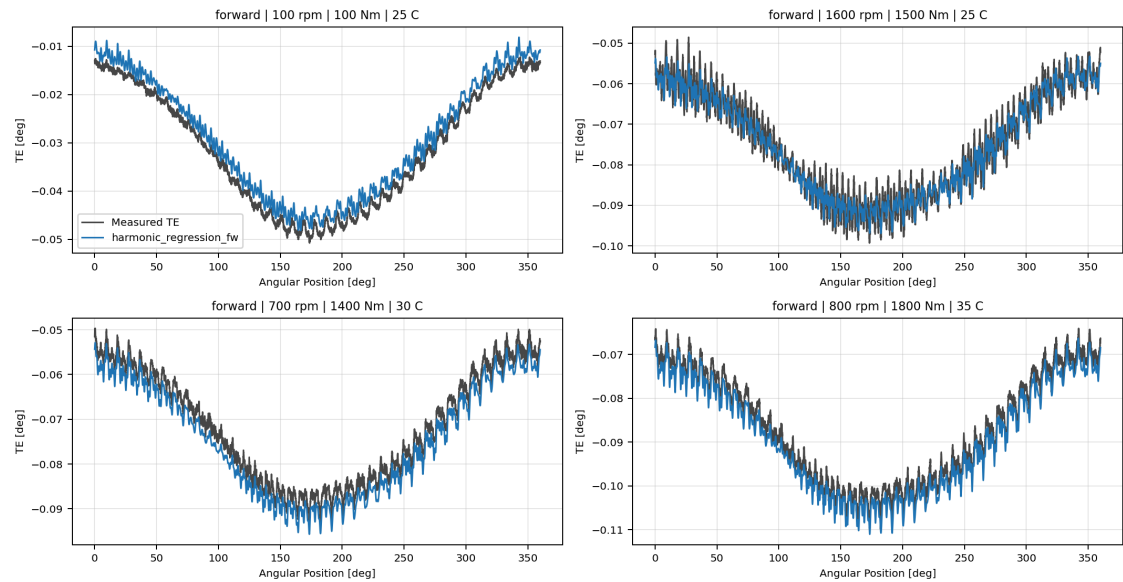
feedforward_fw



feedforward_fw Track 2 collage

harmonic_regression_fw

harmonic_regression_fw

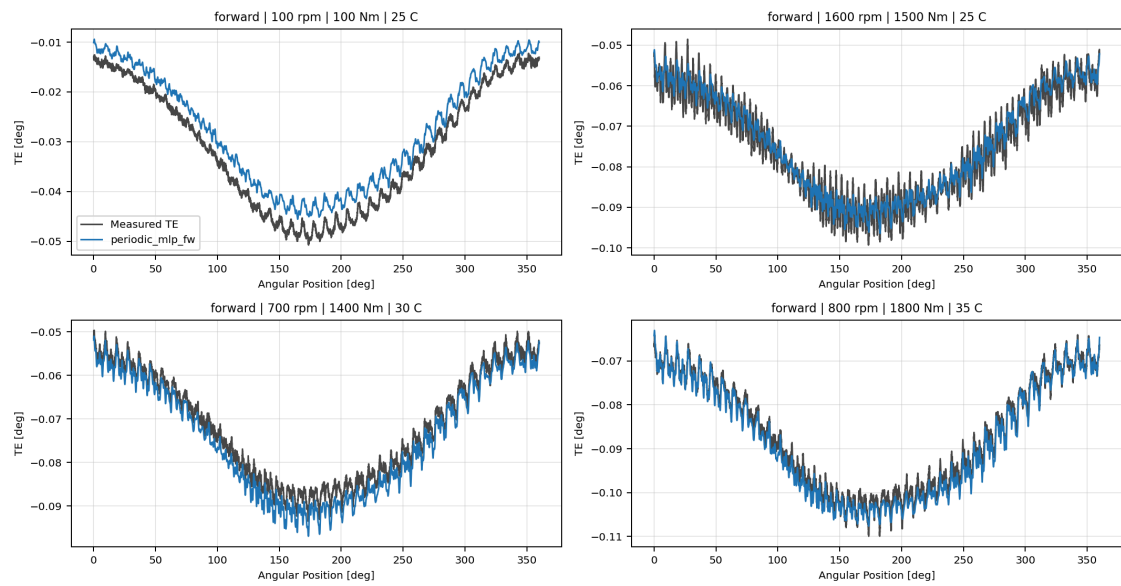


harmonic_regression_fw Track 2 collage

Collage Gallery - Forward Wave 1 Family Best Models Continued

periodic_mlp_fw

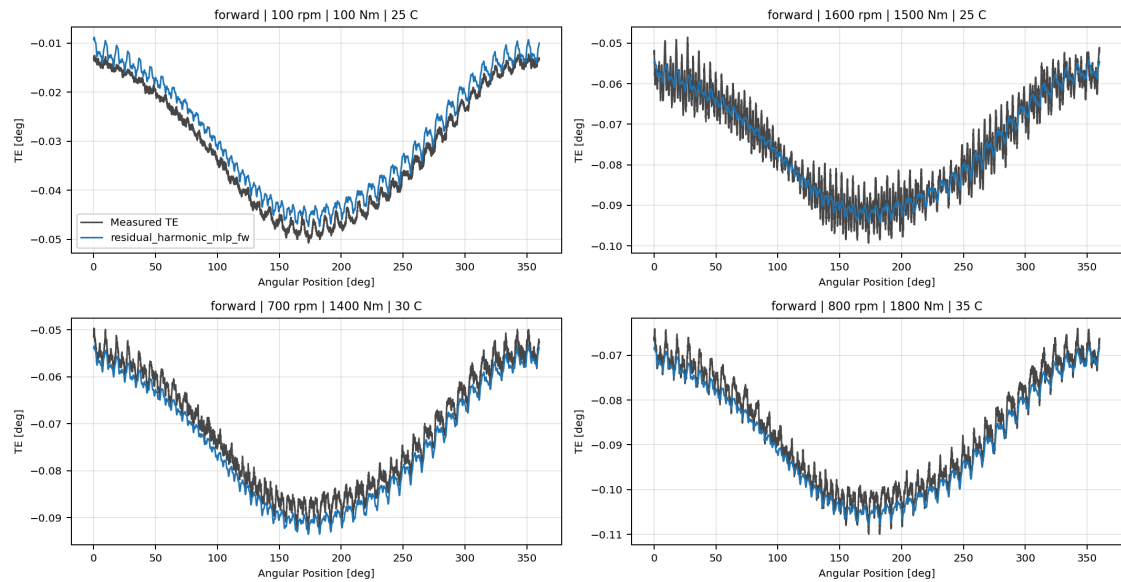
periodic_mlp_fw



periodic_mlp_fw Track 2 collage

residual_harmonic_mlp_fw

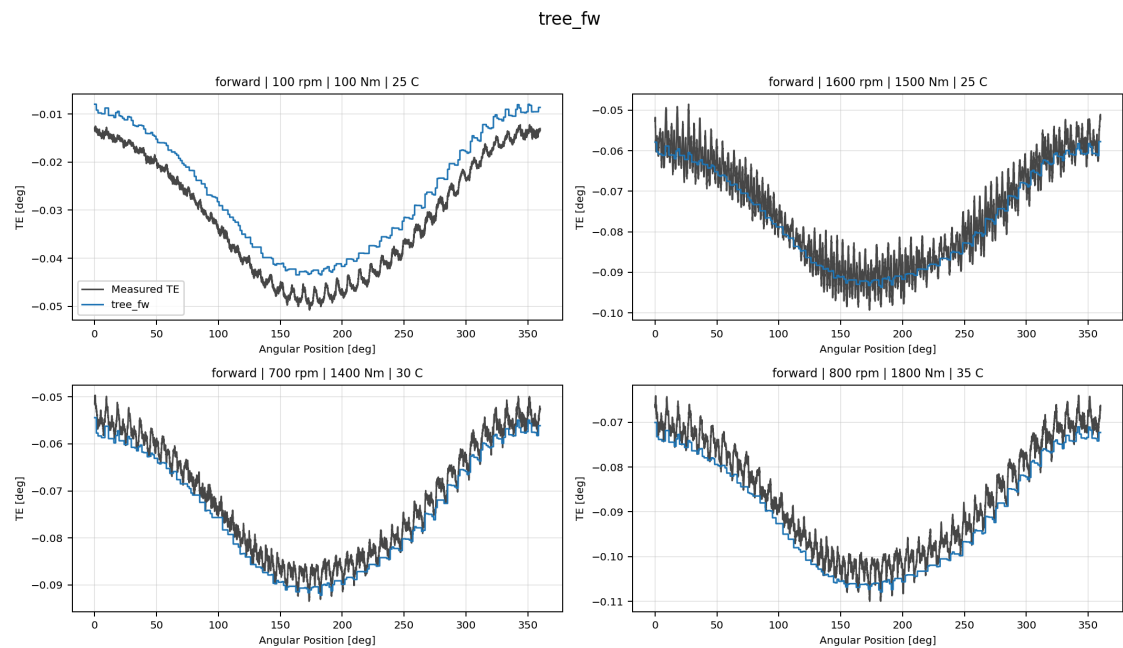
residual_harmonic_mlp_fw



residual_harmonic_mlp_fw Track 2 collage

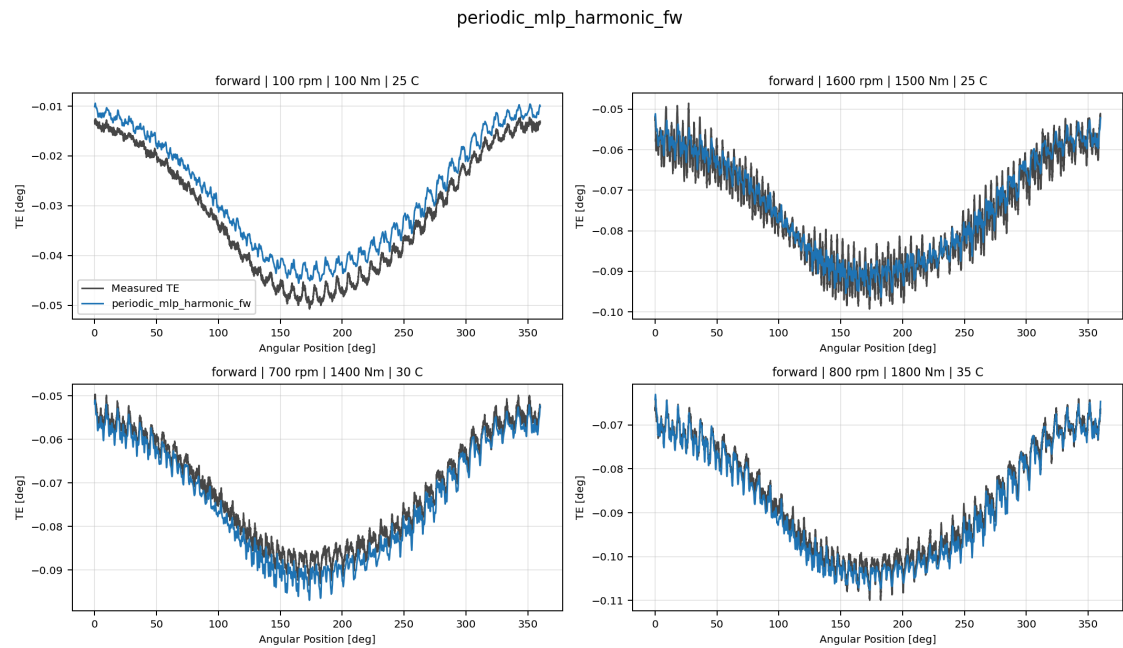
Collage Gallery - Forward Wave 1 Family Best Models Continued 2

tree_fw



tree_fw Track 2 collage

periodic_mlp_harmonic_fw

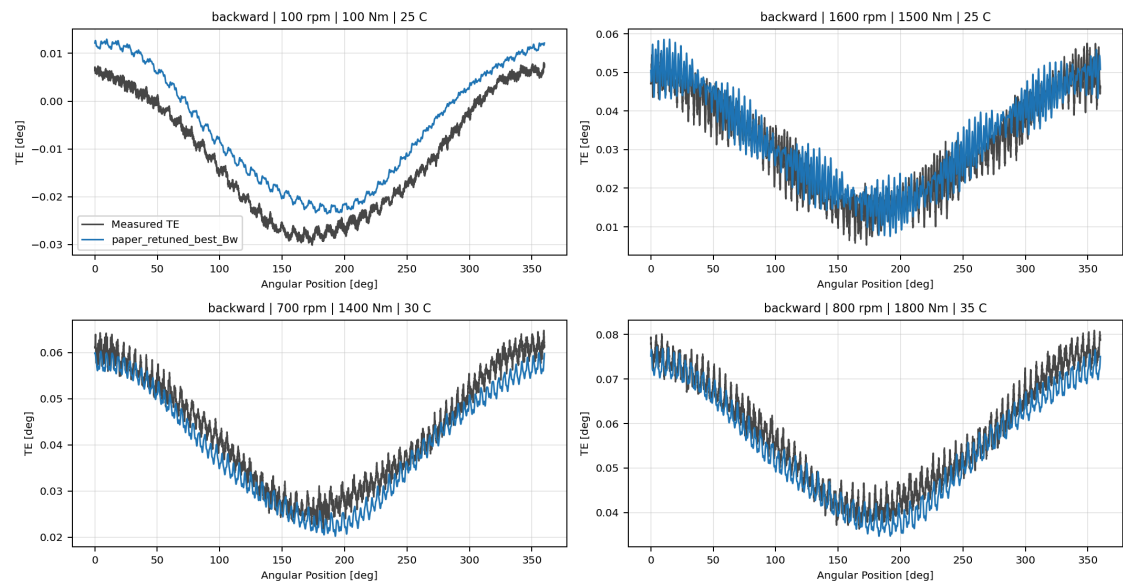


periodic_mlp_harmonic_fw Track 2 collage

Collage Gallery - Backward Reference Best Models

paper_retuned_best_Bw

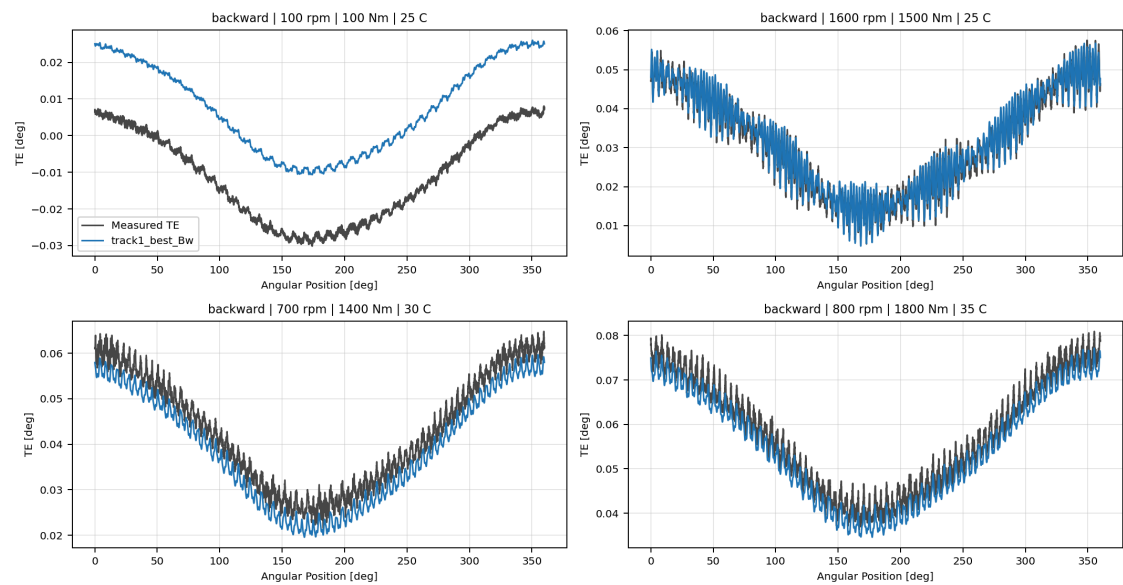
paper_retuned_best_Bw



paper_retuned_best_Bw Track 2 collage

track1_best_Bw

track1_best_Bw

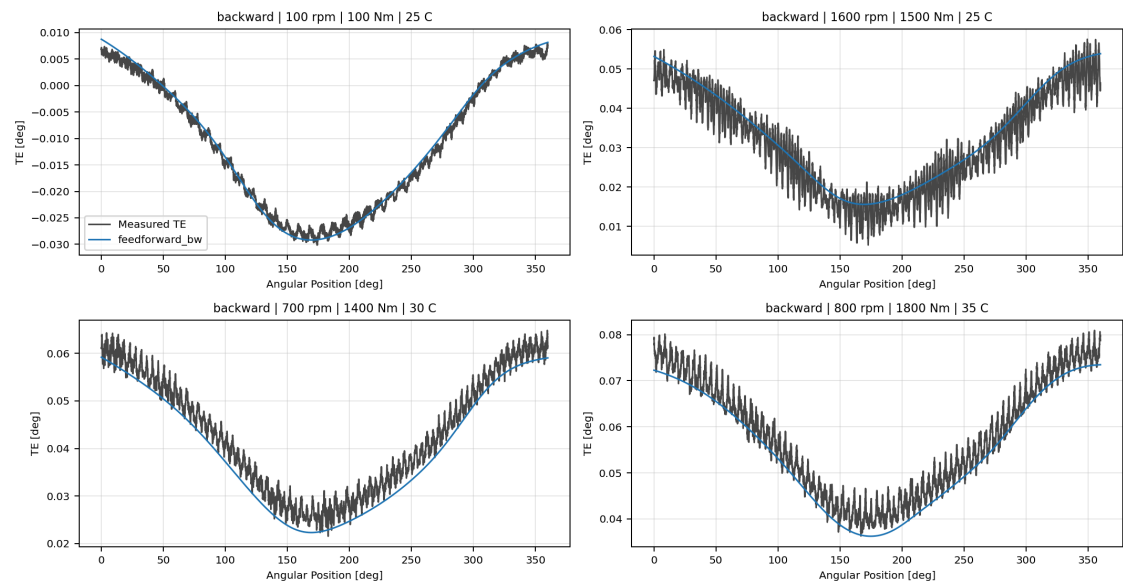


track1_best_Bw Track 2 collage

Collage Gallery - Backward Wave 1 Family Best Models

feedforward_bw

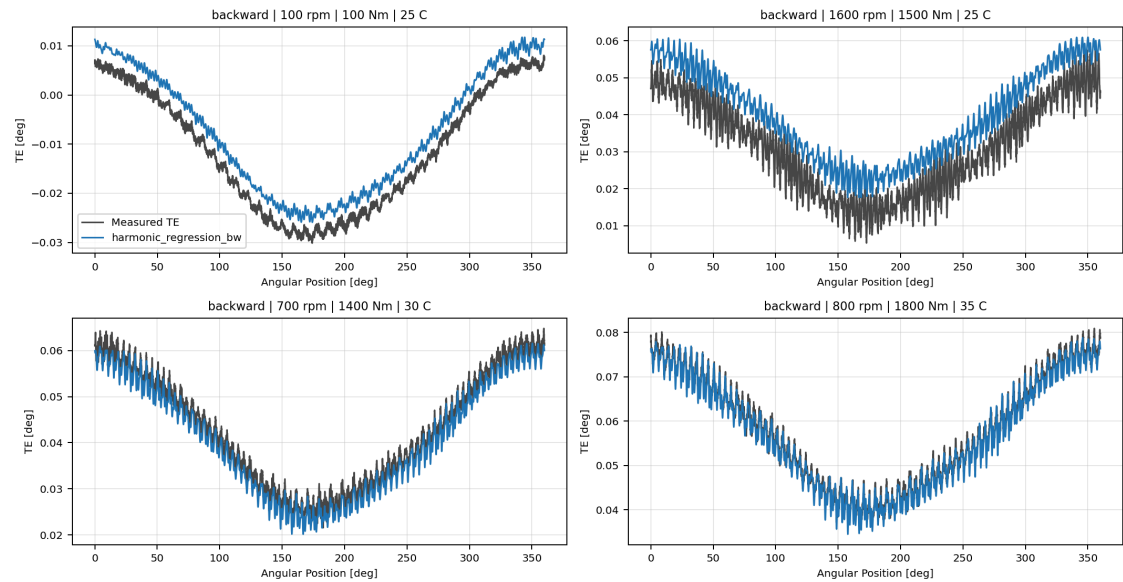
feedforward_bw



feedforward_bw Track 2 collage

harmonic_regression_bw

harmonic_regression_bw

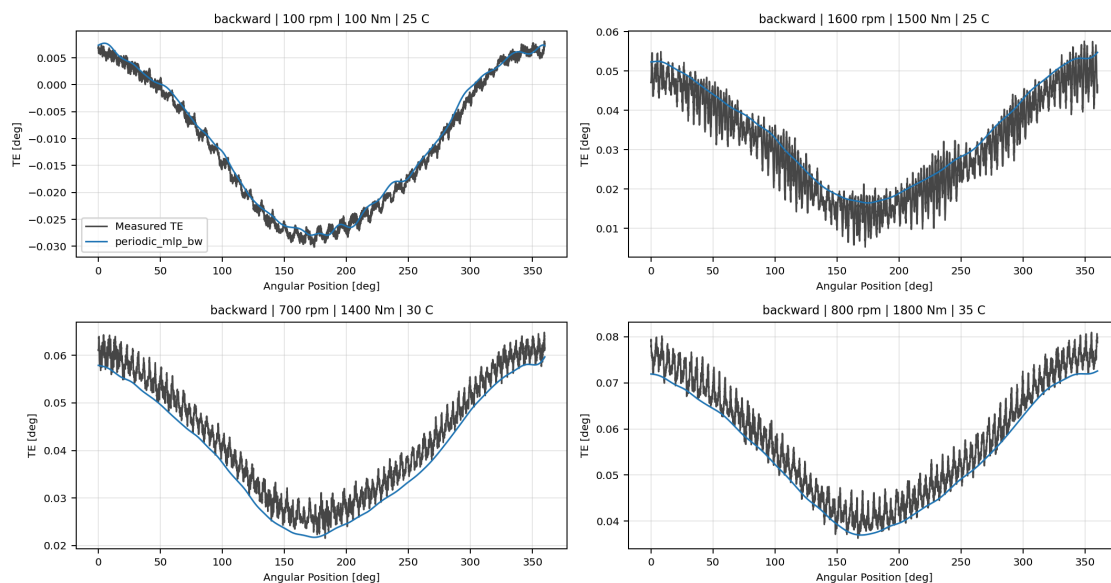


harmonic_regression_bw Track 2 collage

Collage Gallery - Backward Wave 1 Family Best Models Continued

periodic_mlp_bw

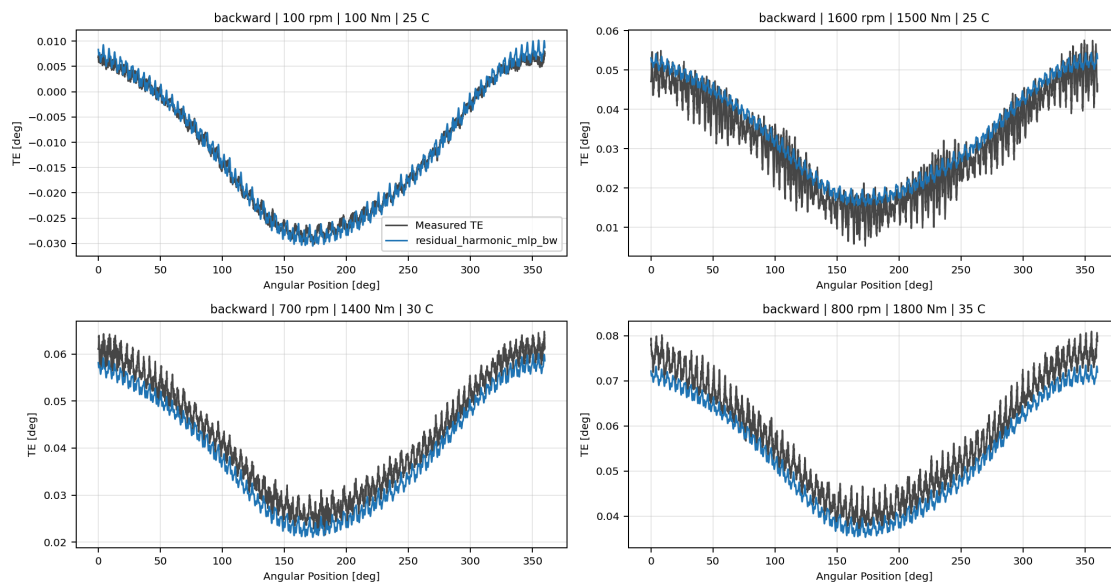
periodic_mlp_bw



periodic_mlp_bw Track 2 collage

residual_harmonic_mlp_bw

residual_harmonic_mlp_bw

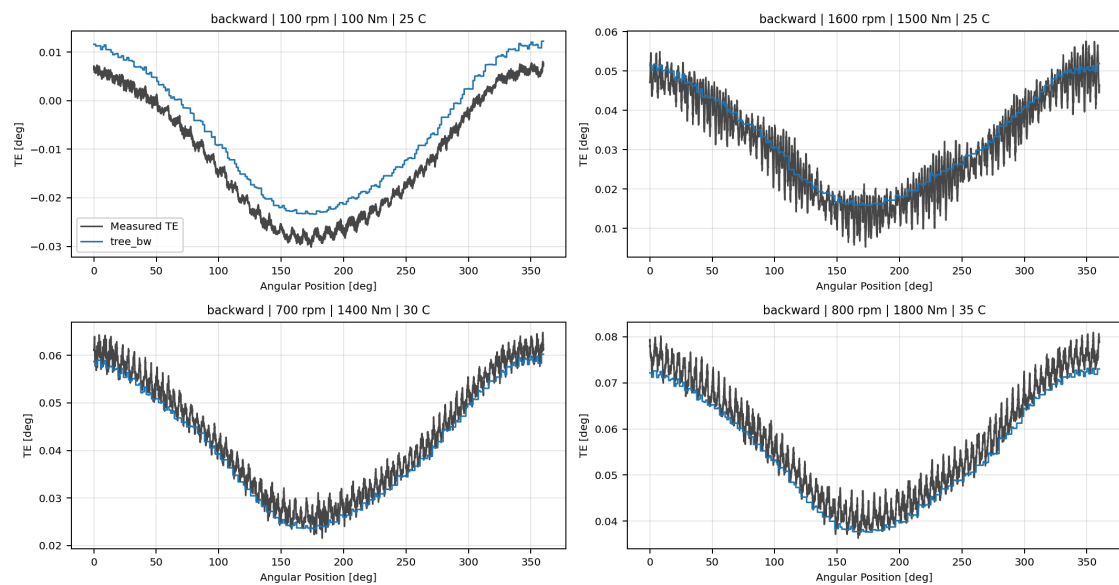


residual_harmonic_mlp_bw Track 2 collage

Collage Gallery - Backward Wave 1 Family Best Models Continued 2

tree_bw

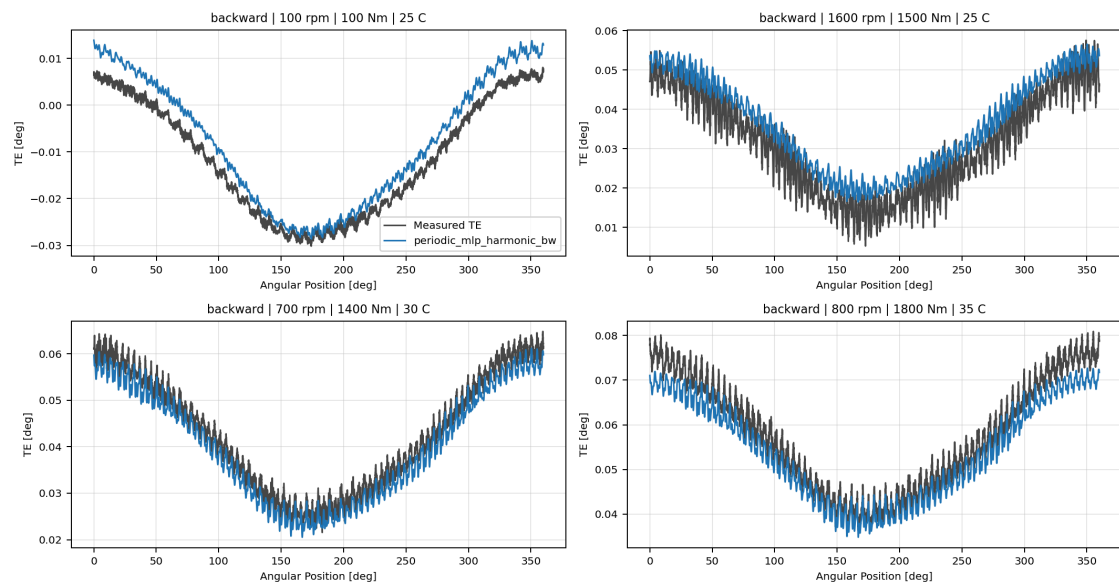
tree_bw



tree_bw Track 2 collage

periodic_mlp_harmonic_bw

periodic_mlp_harmonic_bw

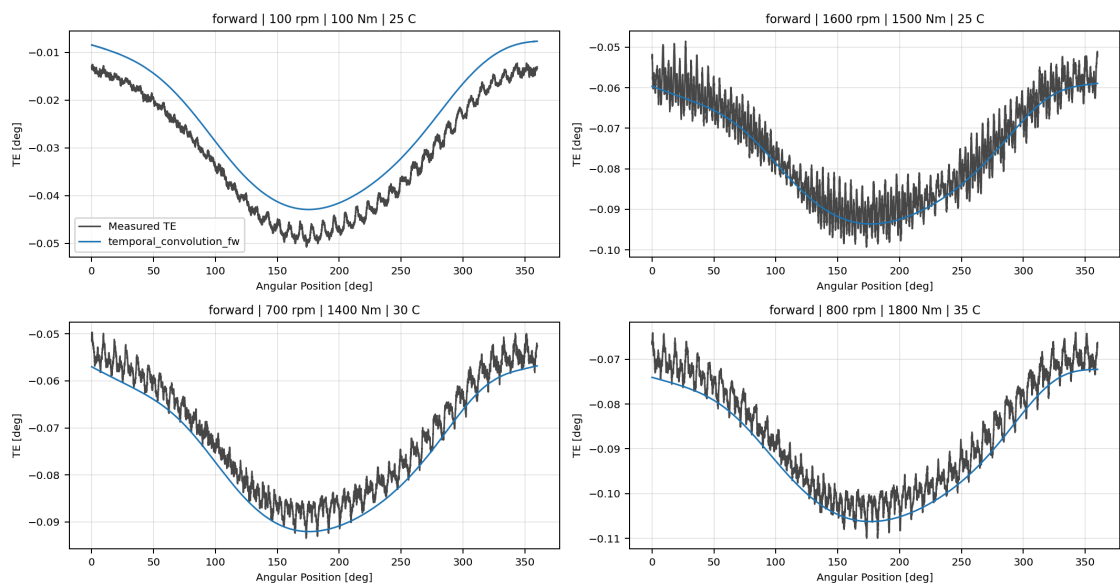


periodic_mlp_harmonic_bw Track 2 collage

Collage Gallery - Forward Wave 2 Temporal Family Best Models

temporal_convolution_fw

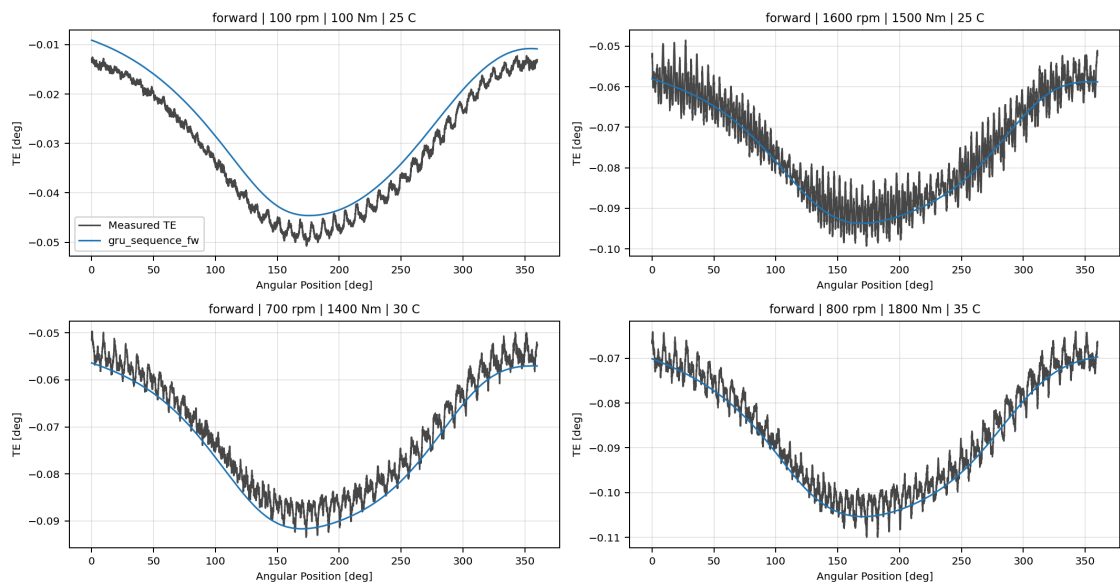
temporal_convolution_fw



temporal_convolution_fw Track 2 collage

gru_sequence_fw

gru_sequence_fw

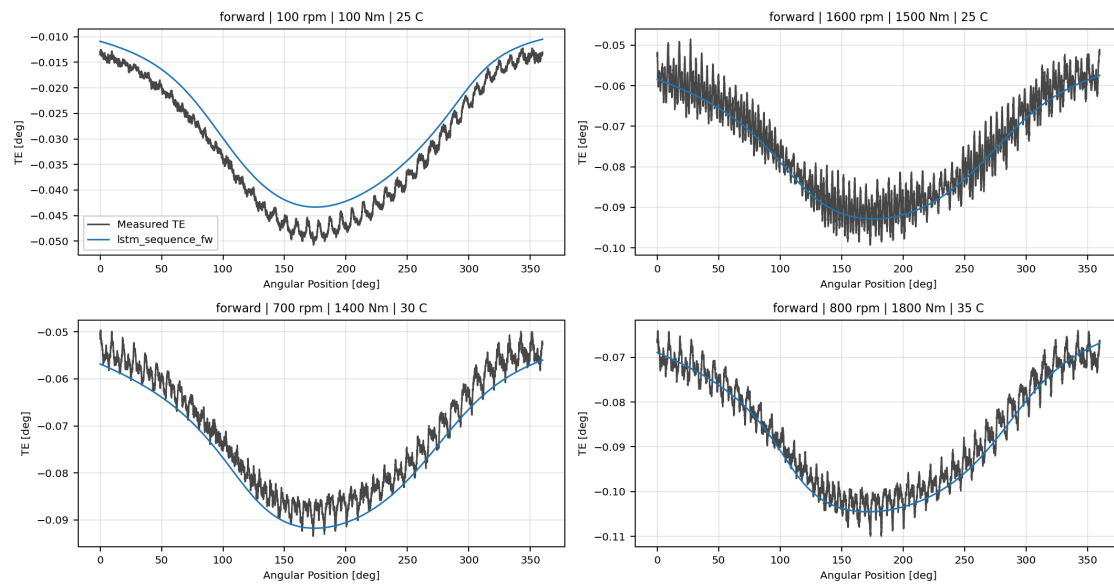


gru_sequence_fw Track 2 collage

Collage Gallery - Forward Wave 2 Temporal Family Best Models Continued

lstm_sequence_fw

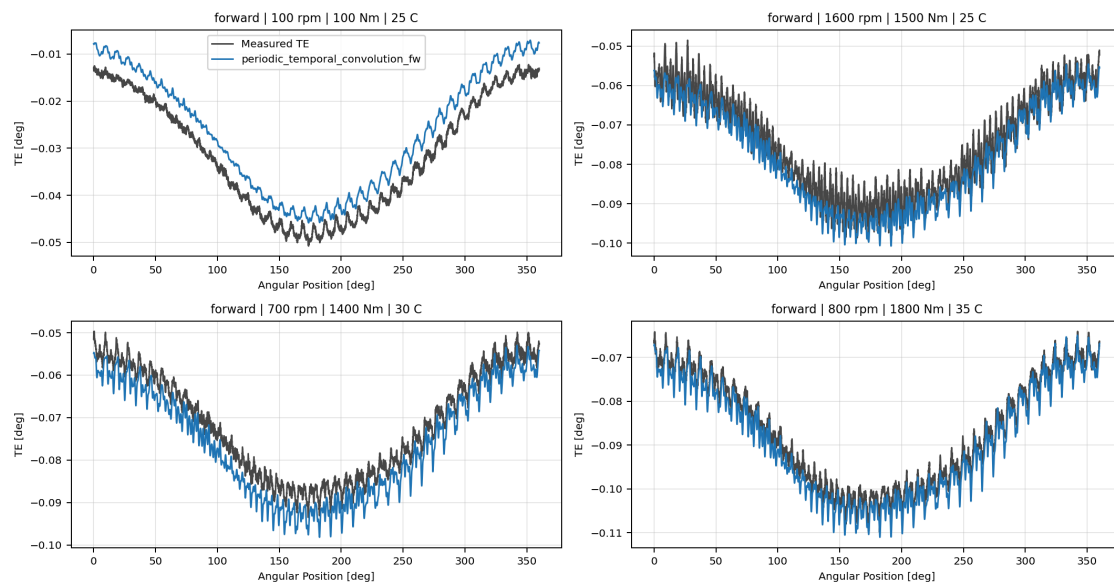
lstm_sequence_fw



lstm_sequence_fw Track 2 collage

periodic_temporal_convolution_fw

periodic_temporal_convolution_fw

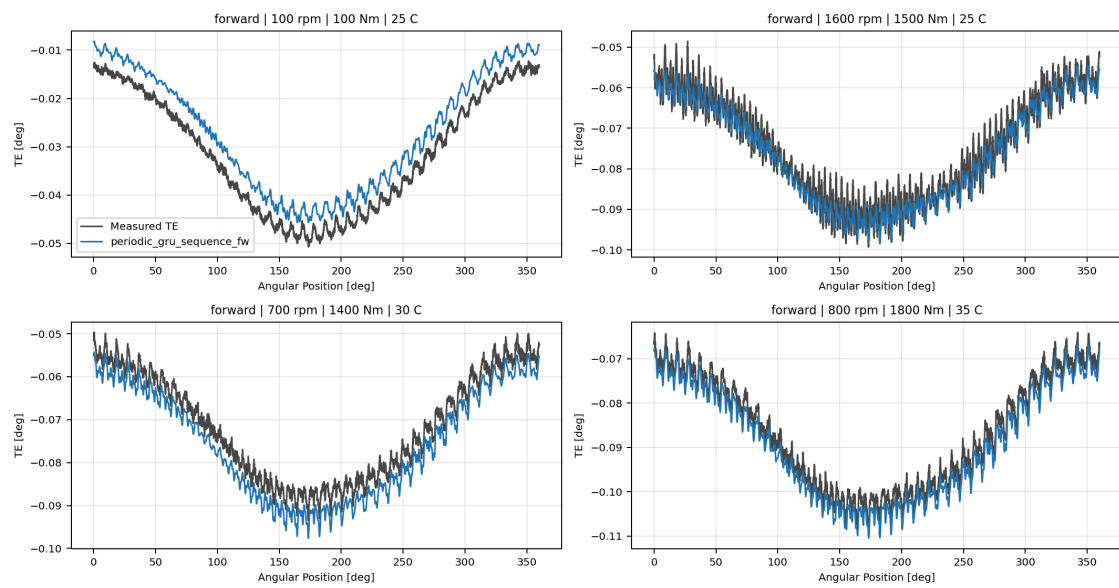


periodic_temporal_convolution_fw Track 2 collage

Collage Gallery - Forward Wave 2 Temporal Family Best Models Continued 2

periodic_gru_sequence_fw

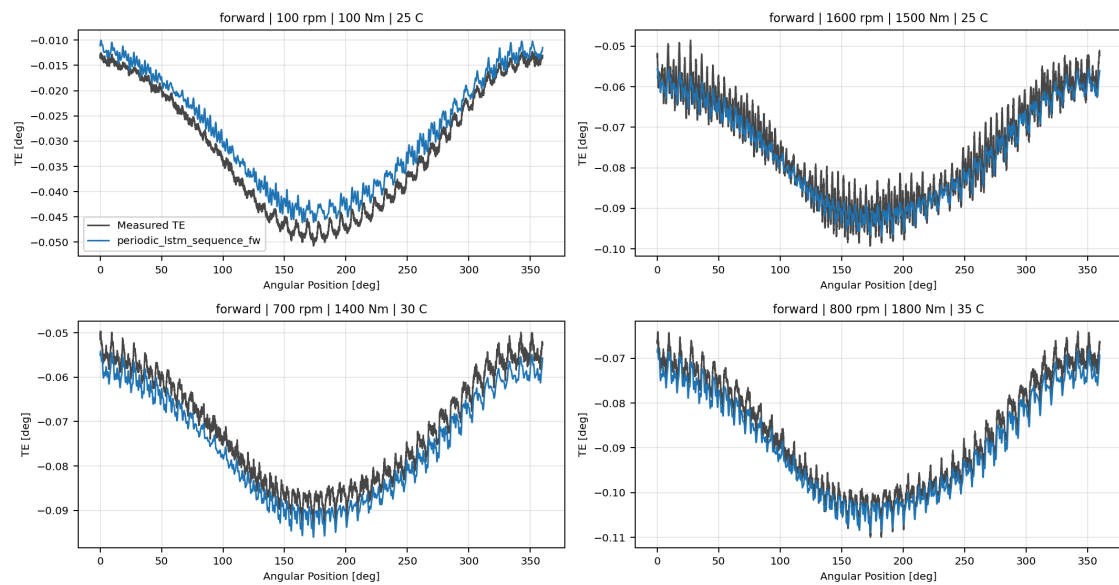
periodic_gru_sequence_fw



periodic_gru_sequence_fw Track 2 collage

periodic_lstm_sequence_fw

periodic_lstm_sequence_fw

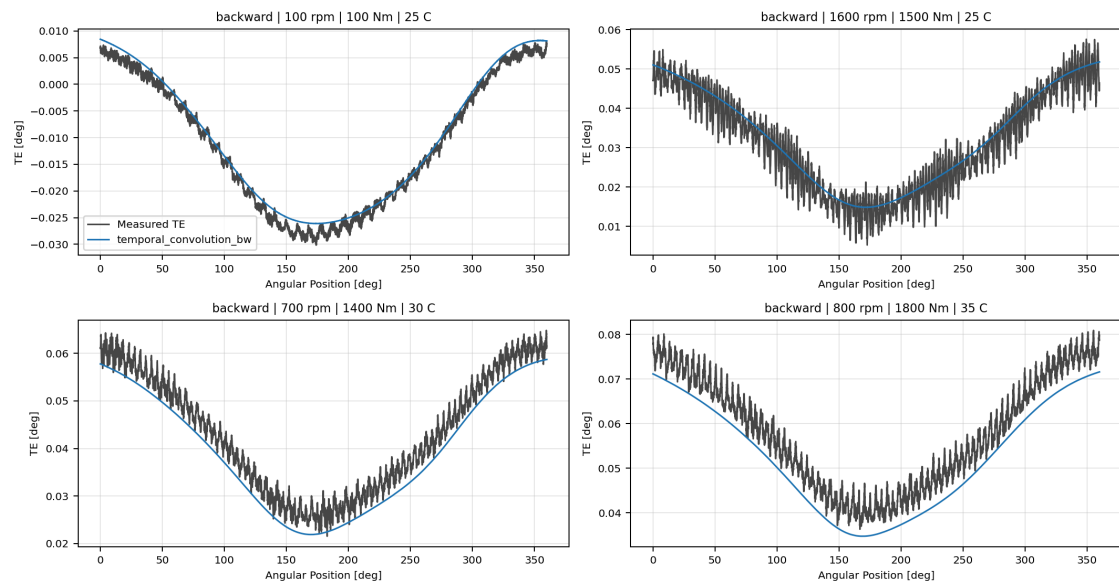


periodic_lstm_sequence_fw Track 2 collage

Collage Gallery - Backward Wave 2 Temporal Family Best Models

temporal_convolution_bw

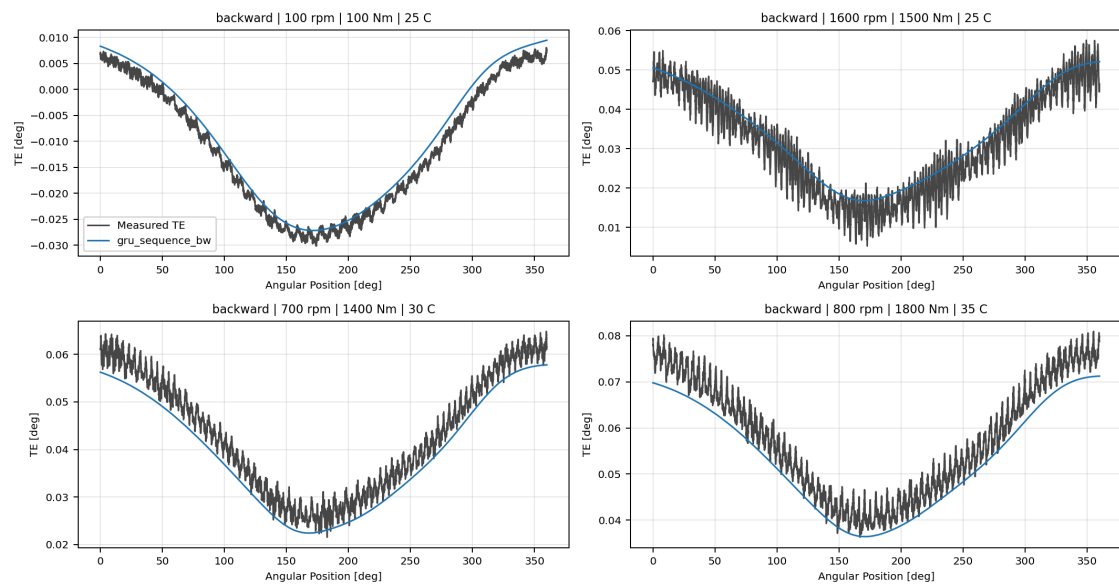
temporal_convolution_bw



temporal_convolution_bw Track 2 collage

gru_sequence_bw

gru_sequence_bw

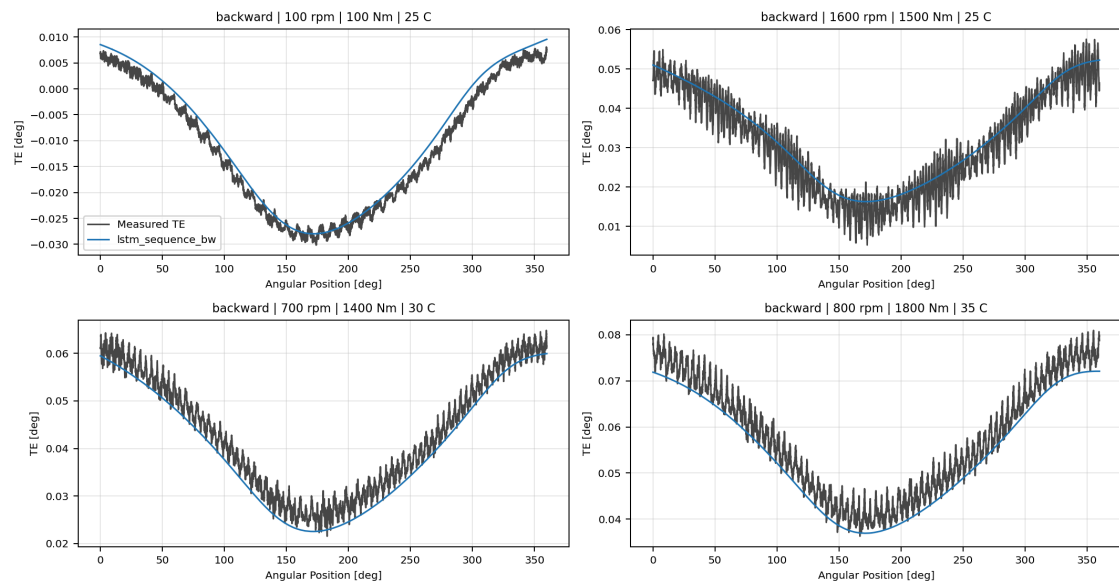


gru_sequence_bw Track 2 collage

Collage Gallery - Backward Wave 2 Temporal Family Best Models Continued

lstm_sequence_bw

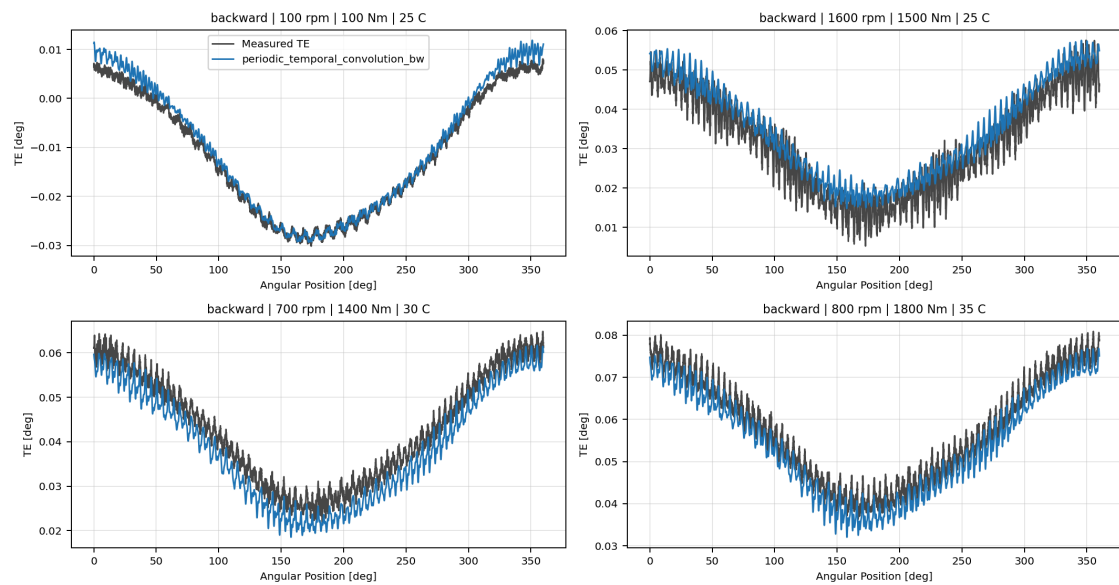
lstm_sequence_bw



lstm_sequence_bw Track 2 collage

periodic_temporal_convolution_bw

periodic_temporal_convolution_bw

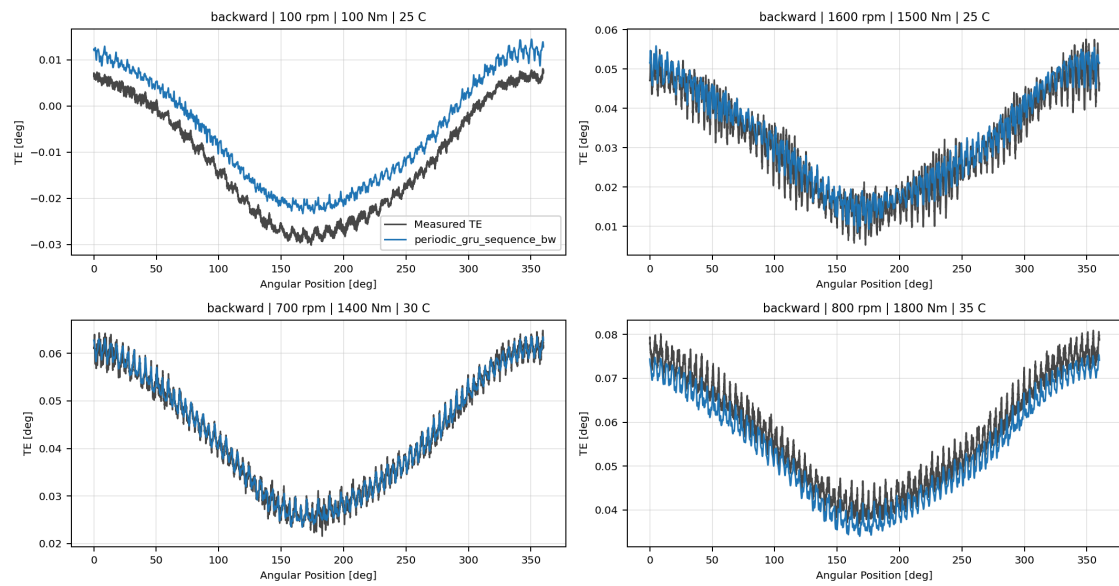


periodic_temporal_convolution_bw Track 2 collage

Collage Gallery - Backward Wave 2 Temporal Family Best Models Continued 2

periodic_gru_sequence_bw

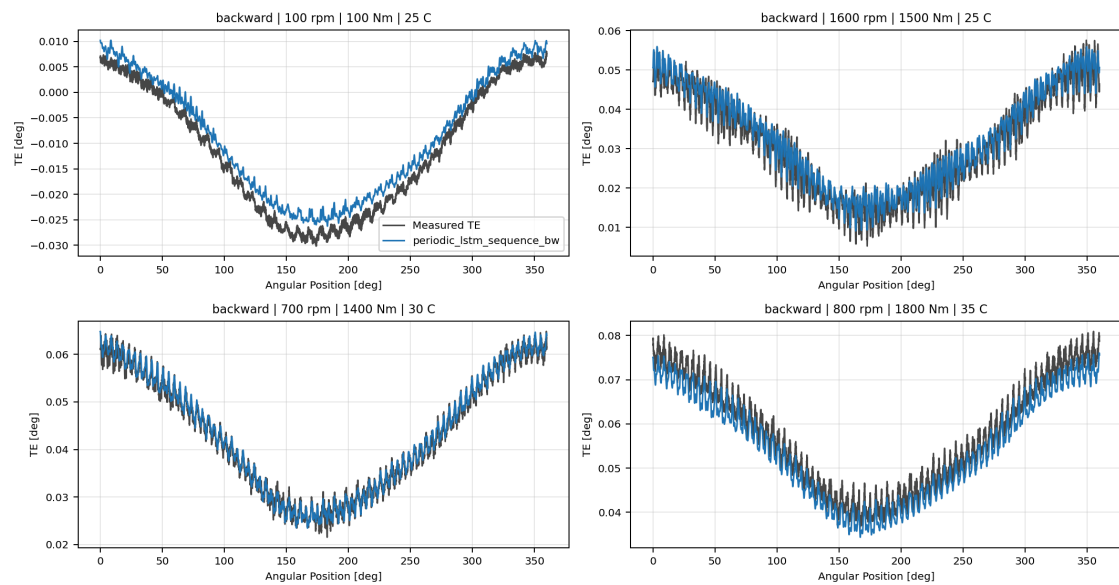
periodic_gru_sequence_bw



periodic_gru_sequence_bw Track 2 collage

periodic_lstm_sequence_bw

periodic_lstm_sequence_bw

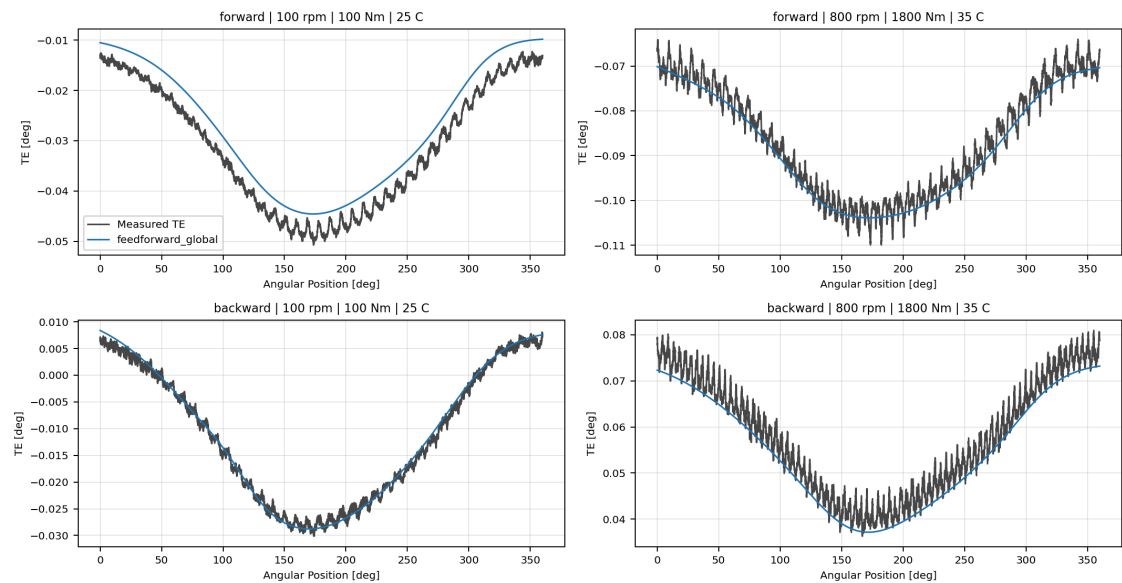


periodic_lstm_sequence_bw Track 2 collage

Collage Gallery - Global Wave 1 Family Best Models

feedforward_global

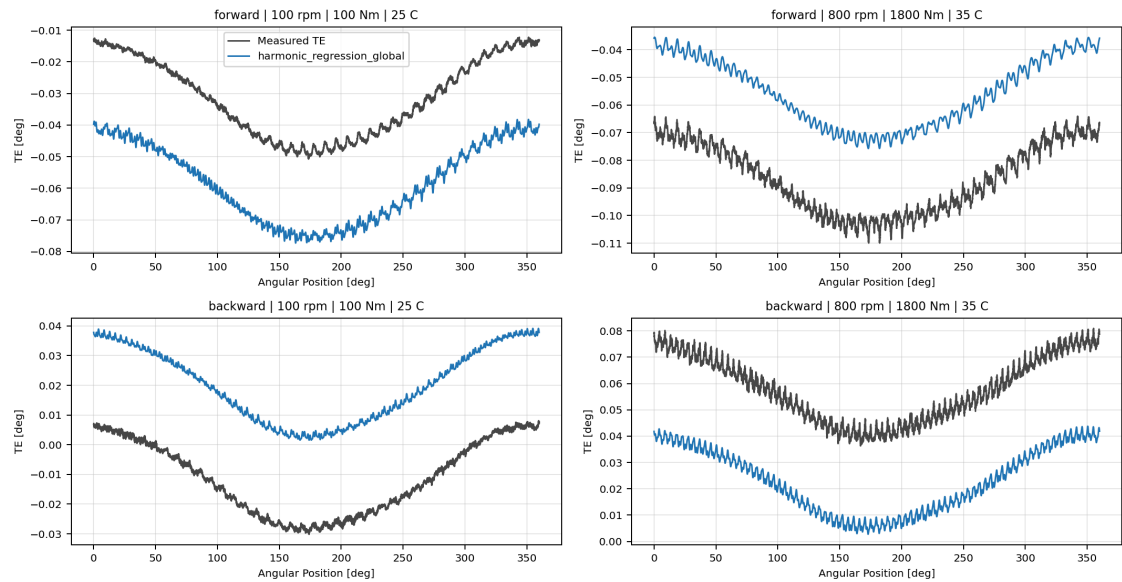
feedforward_global



feedforward_global Track 2 collage

harmonic_regression_global

harmonic_regression_global

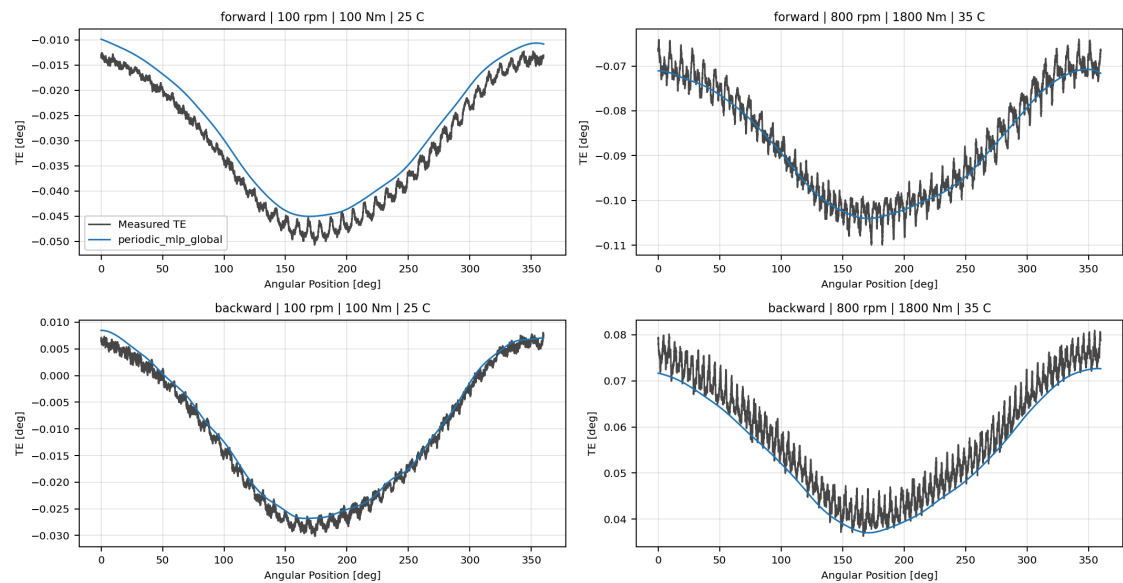


harmonic_regression_global Track 2 collage

Collage Gallery - Global Wave 1 Family Best Models Continued

periodic_mlp_global

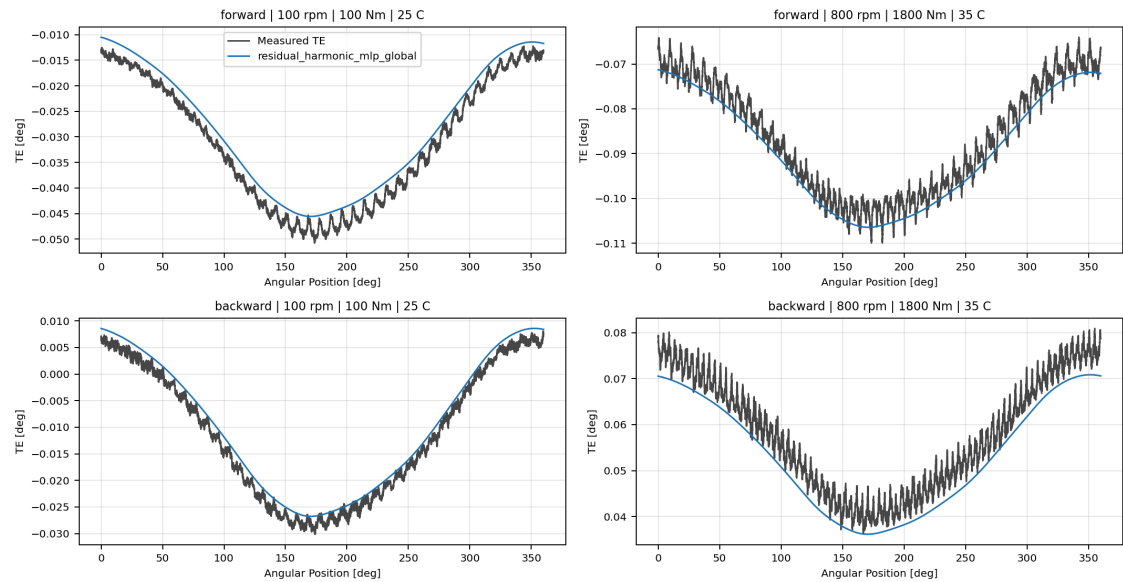
periodic_mlp_global



periodic_mlp_global Track 2 collage

residual_harmonic_mlp_global

residual_harmonic_mlp_global

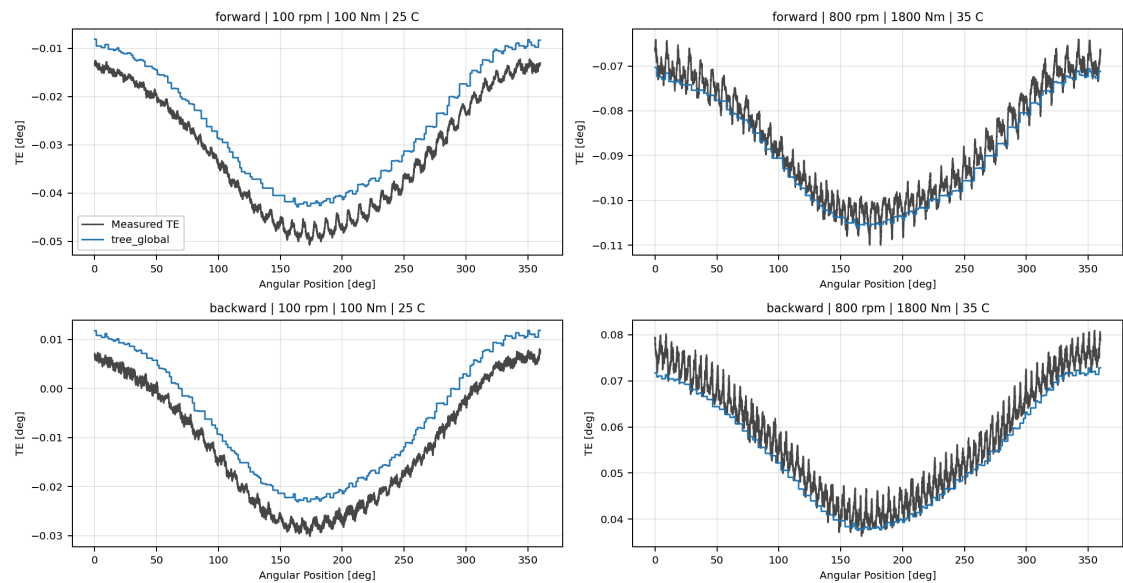


residual_harmonic_mlp_global Track 2 collage

Collage Gallery - Global Wave 1 Family Best Models Continued 2

tree_global

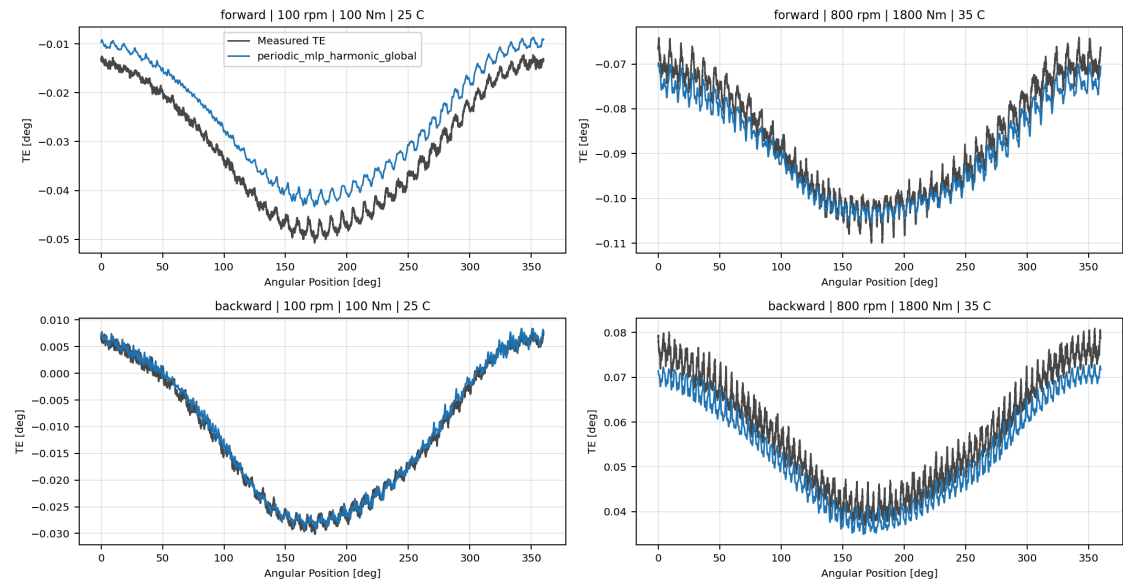
tree_global



tree_global Track 2 collage

periodic_mlp_harmonic_global

periodic_mlp_harmonic_global

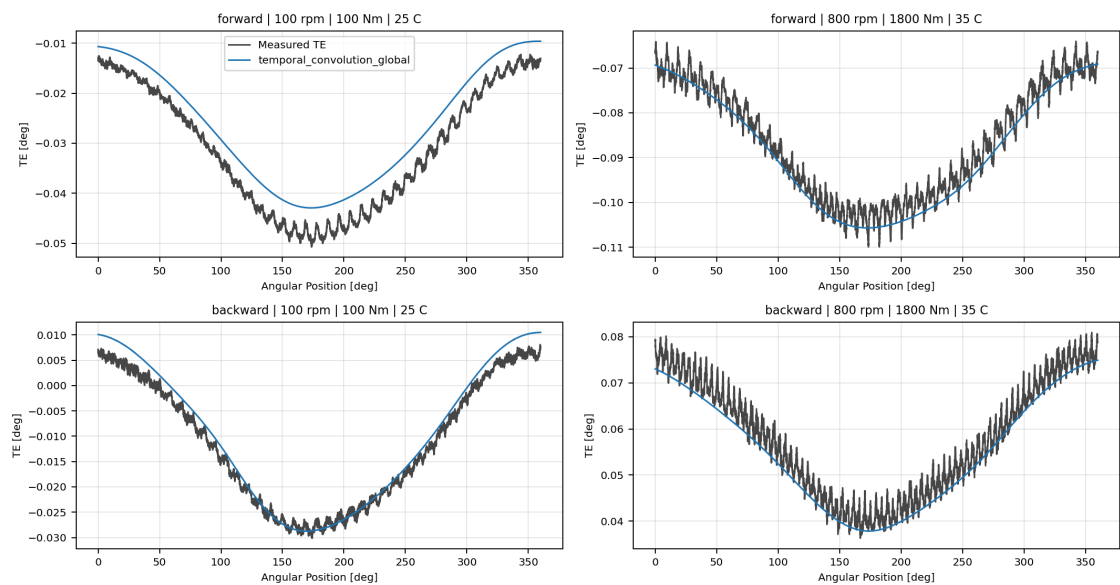


periodic_mlp_harmonic_global Track 2 collage

Collage Gallery - Global Wave 2 Temporal Family Best Models

temporal_convolution_global

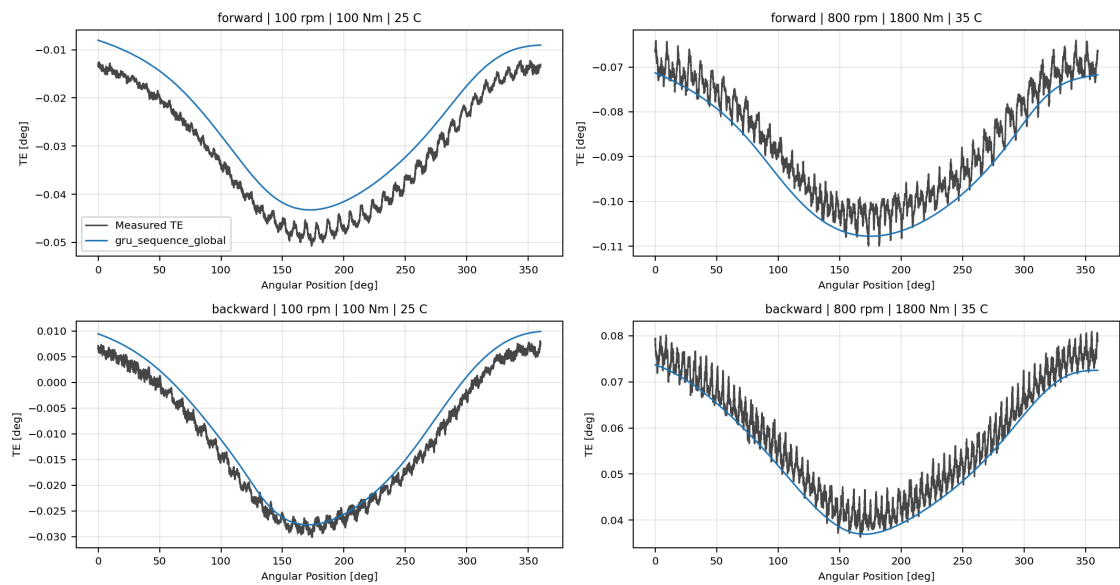
temporal_convolution_global



temporal_convolution_global Track 2 collage

gru_sequence_global

gru_sequence_global

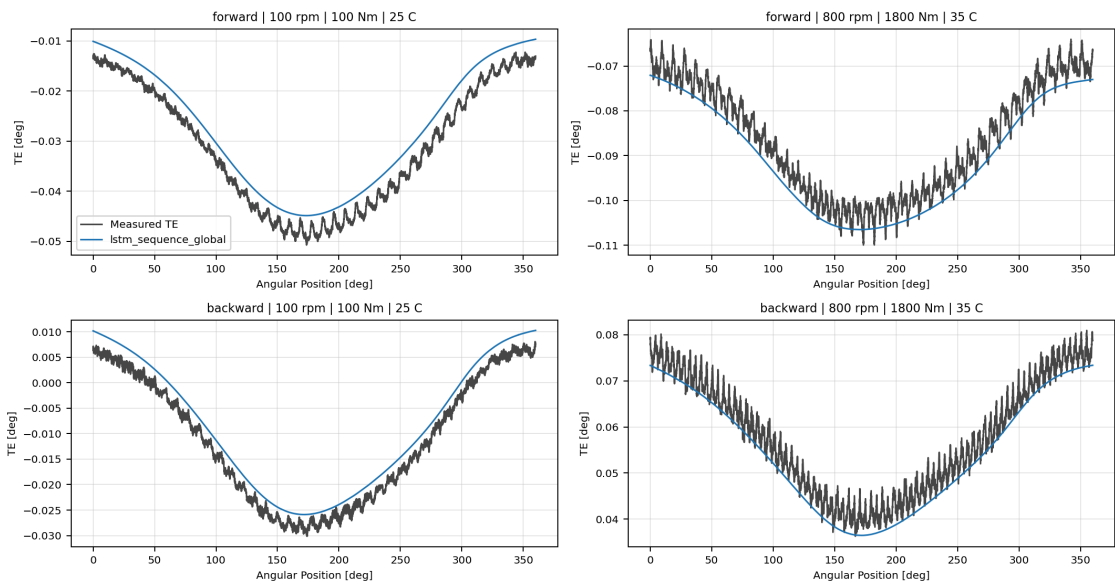


gru_sequence_global Track 2 collage

Collage Gallery - Global Wave 2 Temporal Family Best Models Continued

lstm_sequence_global

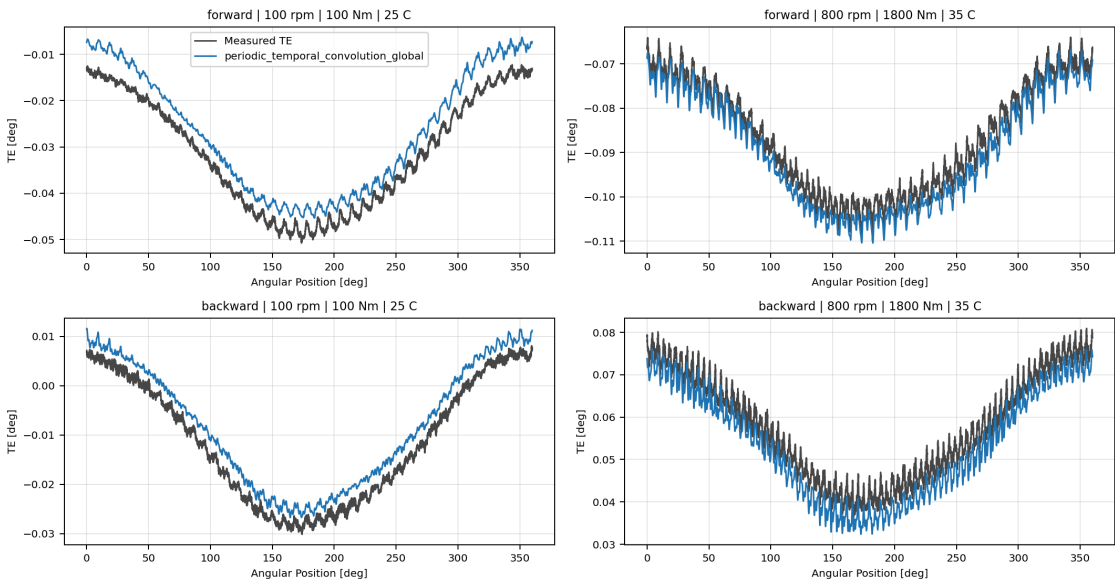
lstm_sequence_global



lstm_sequence_global Track 2 collage

periodic_temporal_convolution_global

periodic_temporal_convolution_global

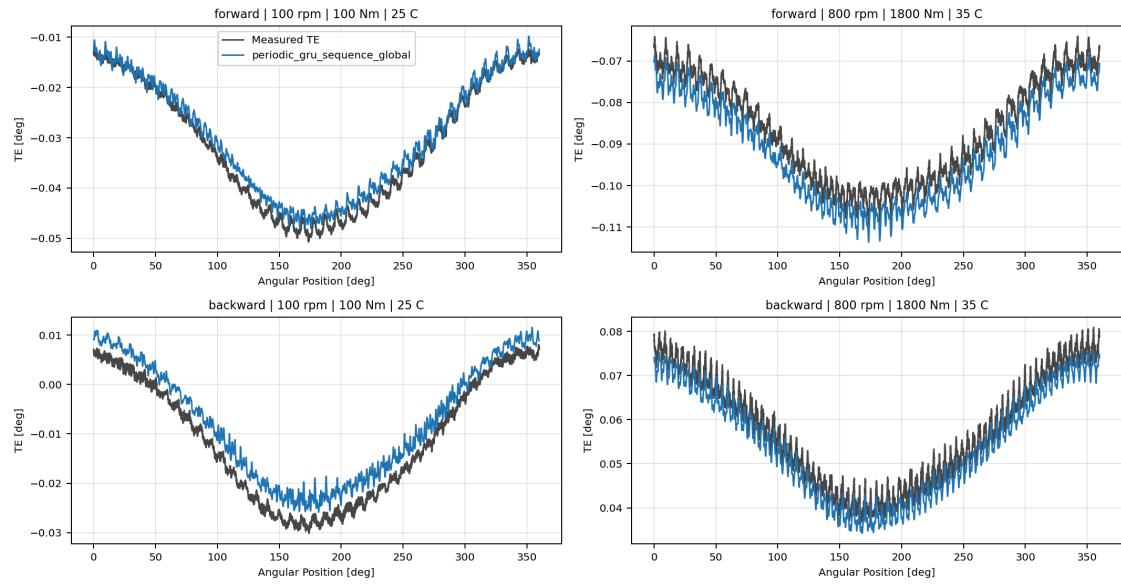


periodic_temporal_convolution_global Track 2 collage

Collage Gallery - Global Wave 2 Temporal Family Best Models Continued 2

[periodic_gru_sequence_global](#)

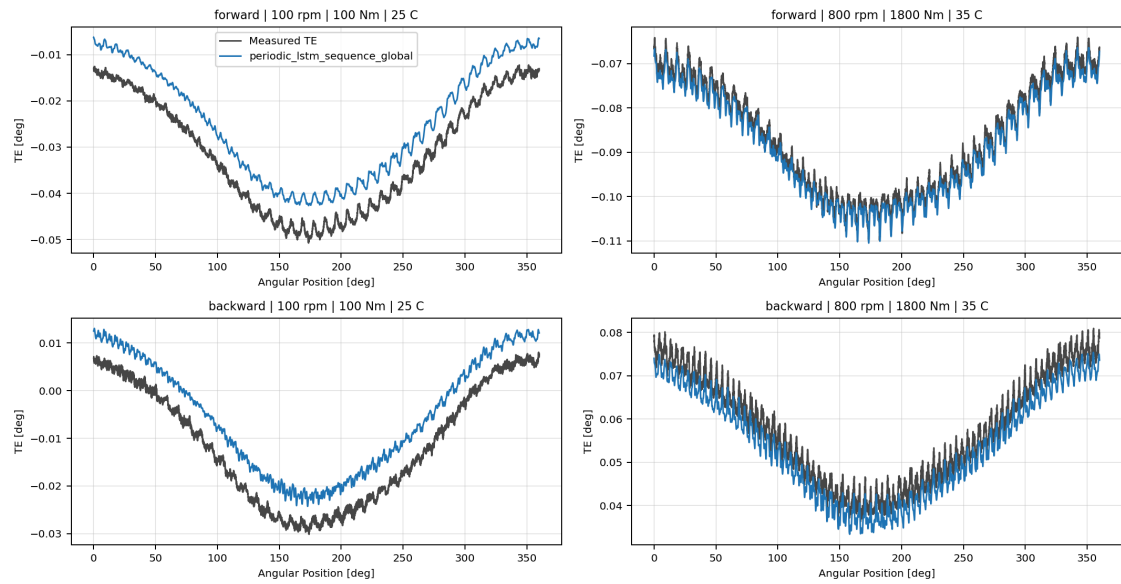
periodic_gru_sequence_global



periodic_gru_sequence_global Track 2 collage

[periodic_lstm_sequence_global](#)

periodic_lstm_sequence_global



periodic_lstm_sequence_global Track 2 collage

Output Artifacts

- output directory: `output\validation_checks\track2_best_model_collage_report\2026-05-27-16-09-04__track2_best_model_collage_report` ;
- summary YAML: `output\validation_checks\track2_best_model_collage_report\2026-05-27-16-09-04__track2_best_model_collage_report\track2_best_model_collage_summary.yaml` ;
- metrics CSV: `output\validation_checks\track2_best_model_collage_report\2026-05-27-16-09-04__track2_best_model_collage_report\track2_best_model_collage_metrics.csv` ;
- report Markdown: `doc\reports\analysis\track2\best_model_collage_report\[2026-05-26]\track2_best_model_collage_report.md` .